

New housing in Prague: permits, completions and the metro

A registered test finds that a permitted flat in a new Prague apartment building takes about 11 months longer to be completed than elsewhere in Czechia (supported), a gap that may partly reflect reporting timing.

PRAGUE, HOMES PER 1 000 RESIDENTS

4.5

a year, 2015–2024 mean · administrative count, no sampling error

PRAGUE METRO REGION, PER 1 000

5.28

a year, 2011–2021 census · 163rd of 208, median 3.55 · administrative count

PERMIT TO COMPLETION, PRAGUE – REST

+11.1 months

bootstrap 90 %: 8.7 to 12.6 · one-sided bound at Holm α : 6.9 · supported; may partly be reporting timing

NEW FLATS, PER DOUBLING OF METRO DISTANCE

–0.23

log density · 90 % Conley: –0.349 to –0.105 · indeterminate: $p = 0.0252$ against 0.025, simulation error about 0.0016

OVERWHELMED? HERE'S THE SHORT VERSION

- Prague completes fewer homes per resident than Vienna or Warsaw, whose metropolitan regions are among Europe's higher builders; by the census count, Prague's own region builds more than most European metropolitan regions.
- The pipeline gap is the one registered test that passed. How long Prague's own lag is remains poorly pinned down: lags from about 14 to 38 months fit almost equally well, while the rest of Czechia's lag is shorter than any block of flats takes to build.
- The gap is measured from the permit to the recorded completion, so it leaves out the time needed to obtain a permit and may partly reflect project size and when completions are registered.
- New flats in Prague were denser near metro stations in the estimate, but the registered test fell 0.0002 short of its threshold and is indeterminate. The European demand model,

the price response and the Polish builders' test are all uninformative, because their relevance checks failed.

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The question and what this adds

Housing economics distinguishes several reasons why a city adds few homes. Demand may be low; procedure may be slow, either in granting permits or in building once a permit is granted; or supply may not respond when prices rise, which happens where regulation binds and prices rather than quantities absorb demand^{2,3,4}. The responsiveness of supply to prices has been estimated from national and metropolitan series^{5,6}, and the Czech discussion of procedure has concentrated on how long a building permit takes, the step the 2021 building act addressed and the one the World Bank's discontinued Doing Business series measured for a stylised warehouse^{7,8}. This study measures the step after the permit with official monthly data, asking how long a permitted flat takes to be completed in Prague compared with the rest of Czechia, and places that measurement beside a census comparison of Prague with Europe's metropolitan regions, a test of where in Prague new flats are built, and a Polish test of whether non-market builders respond differently to prices.

Data

The starting point is a count that each country's statistical office publishes: new homes completed in a year for every thousand residents. Prague has 1.4 million residents, Vienna 2.0 million and Warsaw 1.9 million, and the three series come from the Czech Statistical Office, Statistik Austria and Statistics Poland^{9,10,11,12}.

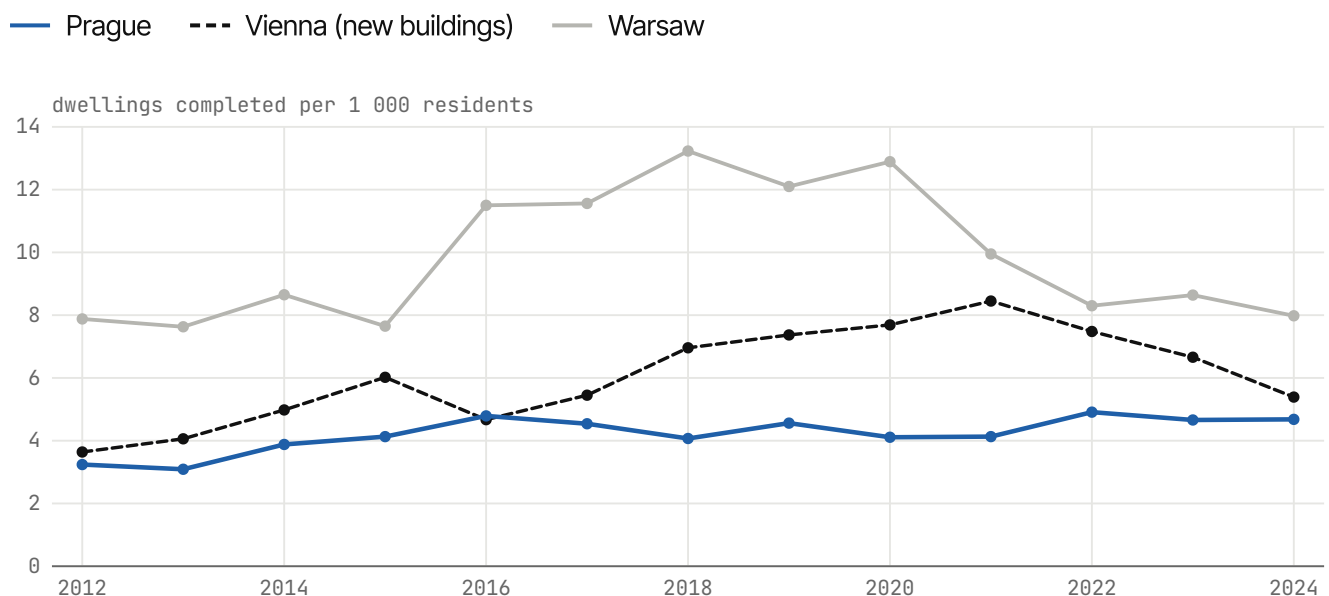


fig. 1 · Dwellings completed per 1 000 residents a year, 2012–2024, in Prague (blue), Vienna (black, dashed) and Warsaw (light grey); administrative counts, no sampling band. Prague and Warsaw count every completed home, including those made by extending or converting a building; Vienna's series counts homes in new buildings only, because conversions were recorded for Vienna only from 2024. With them, Vienna's 2024 figure is 6.2. Sources: ČSÚ, Statistik Austria, Statistics Poland, accessed 28 September 2026.

Prague's line is the flattest of the three and the lowest in every year but 2016. It runs between 3.1 homes per thousand residents in 2013 and 4.9 in 2022, and in 2024 the city completed 6 489 homes. Vienna climbed through the 2010s to 8.5 in 2021, when 16 262 homes were completed in new buildings, and fell to 5.4 in 2024, while Warsaw built more than both in every year, with a peak of 13.2 in 2018, or 23 430 homes. Over the ten years to 2024 Prague averaged 4.5, Vienna 6.6 and Warsaw 10.4. As the European comparison below shows, the

metropolitan regions of both comparators build well above the European median, so these three lines set Prague beside two high builders.

Year	Prague	Vienna, new buildings	Warsaw
2012	3.2 (4 024)	3.6 (6 295)	7.9 (13 496)
2013	3.1 (3 844)	4.1 (7 124)	7.6 (13 128)
2014	3.9 (4 848)	5.0 (8 877)	8.7 (14 964)
2015	4.1 (5 211)	6.0 (10 924)	7.7 (13 306)
2016	4.8 (6 092)	4.7 (8 662)	11.5 (20 119)
2017	4.5 (5 846)	5.5 (10 230)	11.6 (20 331)
2018	4.1 (5 290)	7.0 (13 161)	13.2 (23 430)
2019	4.6 (6 002)	7.4 (14 020)	12.1 (21 599)
2020	4.1 (5 449)	7.7 (14 737)	12.9 (23 543)
2021	4.1 (5 228)	8.4 (16 262)	9.9 (18 528)
2022	4.9 (6 575)	7.5 (14 669)	8.3 (15 465)
2023	4.7 (6 410)	6.7 (13 283)	8.6 (16 092)
2024	4.7 (6 489)	5.4 (10 879)	8.0 (14 873)

Per 1 000 residents, homes completed in brackets. Administrative counts, no sampling error. Data: [cities.json](#).

Each office uses its own definition. Because Vienna's figure leaves out homes made by converting or extending buildings until 2024, Vienna is if anything understated here. Prague's figures are the Czech Statistical Office's own per-thousand rates; its first release for 2024 gave 6 511 completed homes, revised to 6 489 in the current series¹³. Warsaw's rates use Statistics Poland's end-of-year population averaged over consecutive years, revised upward from 2020 after the 2021 census, and they match the office's published rate where one exists (11.50 and 11.56 for 2016 and 2017).

The tests use four further sources. The 2021 census round gives dwellings by the period in which their building was built for Europe's metropolitan regions, and the 2011 round, regional accounts and the metropolitan typology give the covariates^{14, 15, 16, 17, 18, 19, 20}. The Czech Statistical Office publishes permitted and completed dwellings by region every month and a quarterly price index of flats^{21, 22}. For Prague itself, the cadastre's building points give each building's flats and completion date²³, which are placed on the 1 km population grids of 2011 and 2021^{24, 25}. The Polish district panel comes from the Local Data Bank. All raw files are listed with their SHA-256 hashes.

How this was done

The design was written, reviewed and committed to the site's public repository before the results were computed¹: design version 2 at commit 503ef06 on 29 September 2026 at 00:44 and the results at 4e4301c at 01:18, with the branch pushed at 01:38. Commit times are self-reported, so this is a registered analysis plan and not a pre-registration with an independent timestamp. Before registering, the author had already published the three-city comparison and the 2024 builder split (on 28 September 2026), and so knew Prague's approximate completion rate.

Following the review, this text calls the Czech Statistical Office's "zahájené byty", literally dwellings started, permitted dwellings, because they are counted from building permits; the pipeline runs from the permit to the house number that marks completion. Three hypotheses, H2, H4 and H5, form the confirmatory family and share a 5 % family-wise error rate through Holm's procedure²⁶. Two others were demoted to estimates, each with a sentence fixed in advance for what would be said: the European comparison (H1), because Prague's census

numerator was effectively known from the three-city count, and the price response (H3), because with one Prague and 27 quarters no valid test exists at those thresholds. Each question also has a relevance check, which tests whether the model sees the expected relationship where it should; if the check fails, the result is labelled uninformative whatever its p-value.

Power was simulated with sealed scripts that read only series from before the confirmatory windows²⁷. The simulations expected the pipeline test to be decisive at a six-month difference unless monthly noise was as large as the signal, the metro test to be decisive at a 22 % density difference, and the builders' test to be weak. The inference units are the monthly series of two regional aggregates for H2, 242 Polish districts for H4 and 524 one-kilometre squares in Prague's 22 administrative districts for H5.

Registered tests and estimates

Hypothesis	Test	Result
H2 · a permitted Prague home takes longer to get a house number than one in the rest of Czechia	$\mu_P - \mu_R > 0$	supported +11.1 months, $p < 0.001$
H4 · in Poland, municipal and social builders respond less to local price growth than developers	$\beta_{NM} - \beta_M < 0$	uninformative -0.56, $p = 0.17$; relevance check failed
H5 · in Prague, new flats are denser near the metro	$\gamma < 0$	indeterminate -0.23, $p = 0.0252$ against 0.025
H1 · Prague builds less than European peers with its demand estimate, rule fixed in advance	conformal rank	uninformative rank 119 of 208, $p = 0.57$
H3 · Prague's permitting responds less to price growth estimate, rule fixed in advance	90 % EWC interval	uninformative -11 (-86 to +63)

p-values are one-sided. H2 uses a stationary block bootstrap²⁸ with 9 999 draws; H4 a wild score bootstrap²⁹ over 242 Polish districts; H5 the larger of a spatial p-value (Conley³⁰, 1.5 km) and a wild score bootstrap over Prague's 22 administrative districts; H3 equal-weighted cosine (EWC) intervals³¹. Holm thresholds are 0.0167, 0.025 and 0.05. H2 passes the first and H5 misses the second, so testing stops there. H3 is the cumulative response of permitted dwellings, in log points, to a 1 % price rise of flats in each of the twelve preceding quarters, 2018–2024²²; Prague's response relative to the rest of Czechia is -11 (90 % EWC interval -86 to +63), and it is labelled uninformative because the rest's own response, 16 (-22 to +54), cannot be told from zero. Data: [housing_extended.json](#).

Prague among 208 European metropolitan regions

The 2021 census round recorded when each dwelling's building was built. Dwellings in buildings from 2011 or later, per resident and per year, therefore measure the last decade's new supply on nearly the same definition across 22 countries, although Czechia, Spain and Croatia also count reconstructions. The unit is the metropolitan region, because housing markets do not stop at city limits, and Prague's region is Prague plus Středočeský kraj. Of Eurostat's 297 metropolitan regions, 208 have the census answer and the covariates.

Per resident, Prague's metropolitan region built 5.28 dwellings a year by the census count, which places it 163rd of 208 from the bottom, in Europe's upper quarter; the median region built 3.55. Counting only the Czech Statistical Office's completions of new houses gives 3.88, mid-table at 116th of 208 on raw rates against a median of 3.50. Vienna's region built 6.2 a year and Warsaw's 10.0. A descriptive comparison within Czechia, which is not part of the registration, uses the raw Czech Statistical Office rate: by it Prague built the second most

of the 14 Czech regions per resident, 3.7 a year against a median of 2.1 elsewhere. On both counts Prague's region builds more than most European metropolitan regions and less than Vienna's and Warsaw's.

The registered estimate compares each region with a prediction from what was known in 2011: population growth to 2011, size, output per head, dwellings per head, and whether the region holds a capital. The prediction for each country's regions comes from a model fitted on all other countries, and the gap is scaled by how well such regions are usually predicted^{32, 33}.

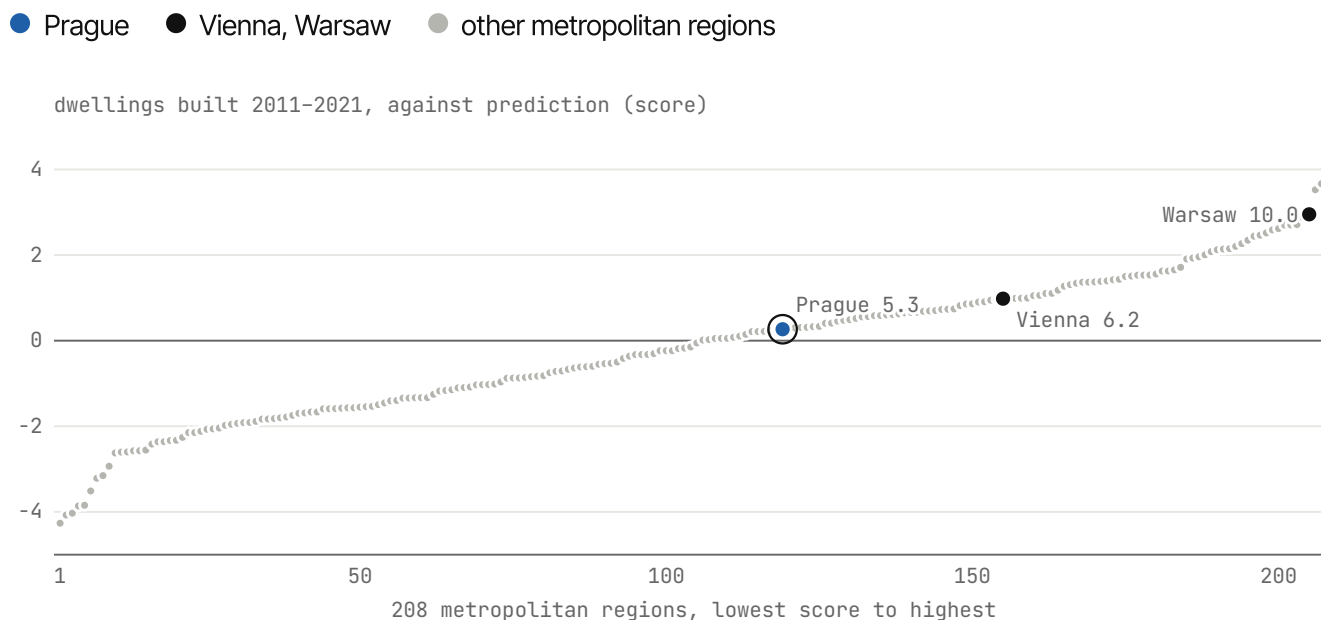


fig. 2 · Dwellings built 2011-2021 against a demand model, 208 European metropolitan regions, studentised leave-one-country-out scores; zero is "as the model predicts". Prague in blue, Vienna and Warsaw in black, other regions in light grey. The model failed its relevance check, so the ranks are shown as description, not as a benchmark. Source: Eurostat census 2021 and 2011, regional accounts, accessed 28-29 September 2026.

Against that model Prague's region ranks 119th from the bottom, Vienna's 155th and Warsaw's 205th. The city on its own ranks 107th, and with the count rebuilt from the Czech Statistical Office's completions of new houses, Prague ranks 93rd as a metro region and 76th as a city. Czechia's mean score is 0.18, 8th of 22 countries from the bottom. The rule written in advance would call this "within the range of European peers", but the model failed its relevance check: regions that grew faster before 2011 did not build more afterwards, and the coefficient is negative. By the same rule the ranks are labelled uninformative.

From permit to completion in Prague and the rest of Czechia

Every month the Czech Statistical Office counts dwellings permitted and dwellings completed in each region. If Prague's flats take longer to finish, completions follow permits with a longer delay. The model estimates that delay as a distribution of lags between the two series, for flats in new apartment buildings, Prague against the other twelve regions, and the estimand is the difference between the two mean lags. Středočeský kraj is left out of the comparison, because it absorbs Prague's overflow.

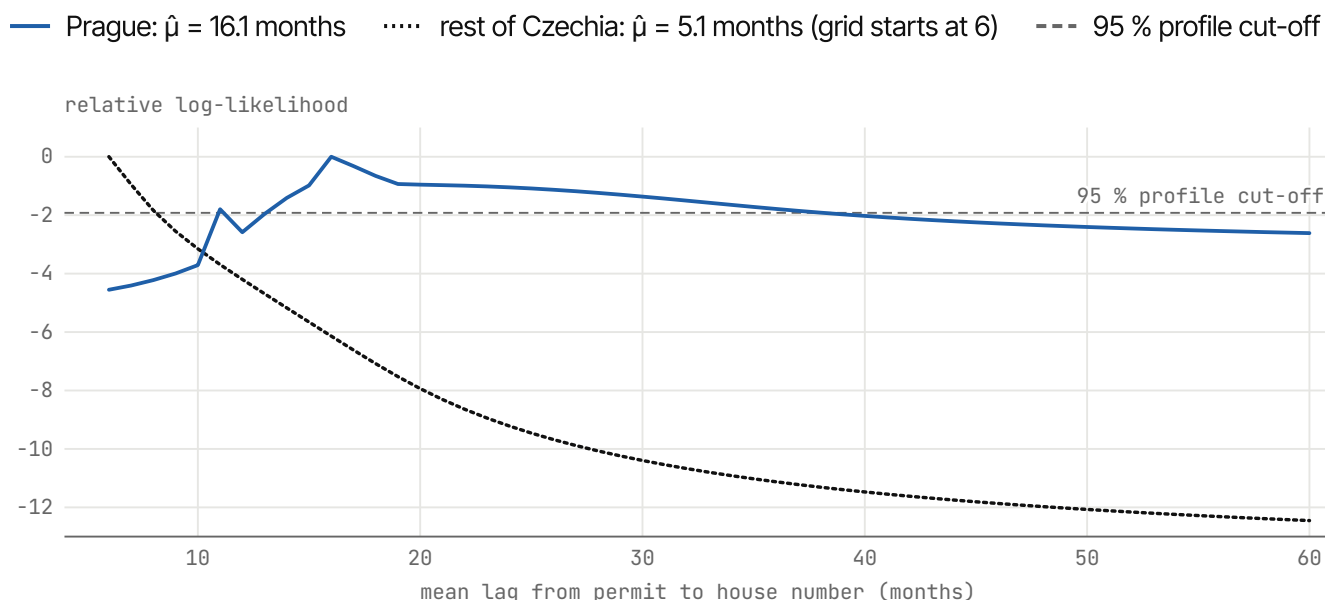


fig. 3 · How well each mean lag fits the data (relative profile log-likelihood), for Prague (blue) and the rest of Czechia without Středočeský kraj (black, dotted), on a grid of 6 to 60 months; monthly completions 2013–2024. A flat curve means the data barely distinguish between lags. Grey dashed: the 95 % cut-off.
Source: ČSÚ, dwellings by region (STA09B), accessed 28 September 2026.

The best-fitting mean lag is 16.1 months in Prague and 5.1 elsewhere, a difference of 11.1 months, and H2 is supported ($p < 0.001$). The p-value uses the bootstrap standard error of 1.93 months with a normal approximation, so the interval that matches the decision is the one-sided lower bound at Holm's threshold of 0.0167, which is 6.9 months. The 5th to 95th percentiles of the bootstrap draws, 8.7 to 12.6 months, are narrower than that standard error implies, because a few draws lie far from the rest.

Fig. 3 shows why both bootstrap figures understate how uncertain Prague's own lag is. The 95 % profile-likelihood region for Prague's mean lag runs from about 14 to 38 months, with one isolated grid point at 11 months also inside it, while the region for the rest of Czechia ends at about 8 months. Prague's fitted lag distribution is degenerate, with all its weight on the single lag of 16 months, and every bootstrap draw is refitted starting from that estimate, so the draws probably stay close to it. Read from the profile likelihood, the two regions do not overlap, so the sign of the difference holds, but its size lies somewhere between about 6 months (3 if the isolated point at 11 months is counted) and more than 30 months. In the bootstrap the completion yield and the lag are almost uncorrelated (-0.05), and no weight falls beyond the six-year window.

What the difference measures also needs qualifying. Five months is too short to build a block of flats, so outside Prague the fit probably also tracks when completions are reported relative to permits, and Prague's projects may be larger, in which case part of the lag reflects project size rather than procedure. The 11 months are therefore a difference in the recorded pipeline, from permit to house number, which may partly be reporting timing, and not a clean difference in construction time. The result also says nothing about how long the permit itself took, which is the part of the process Czech reform has targeted².

Who builds: Vienna, Warsaw and 380 Polish districts

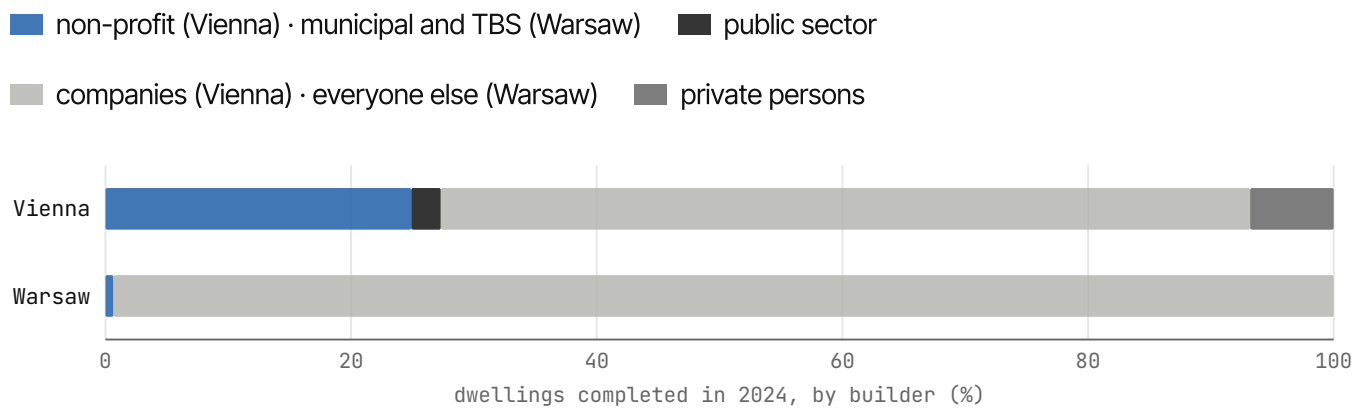


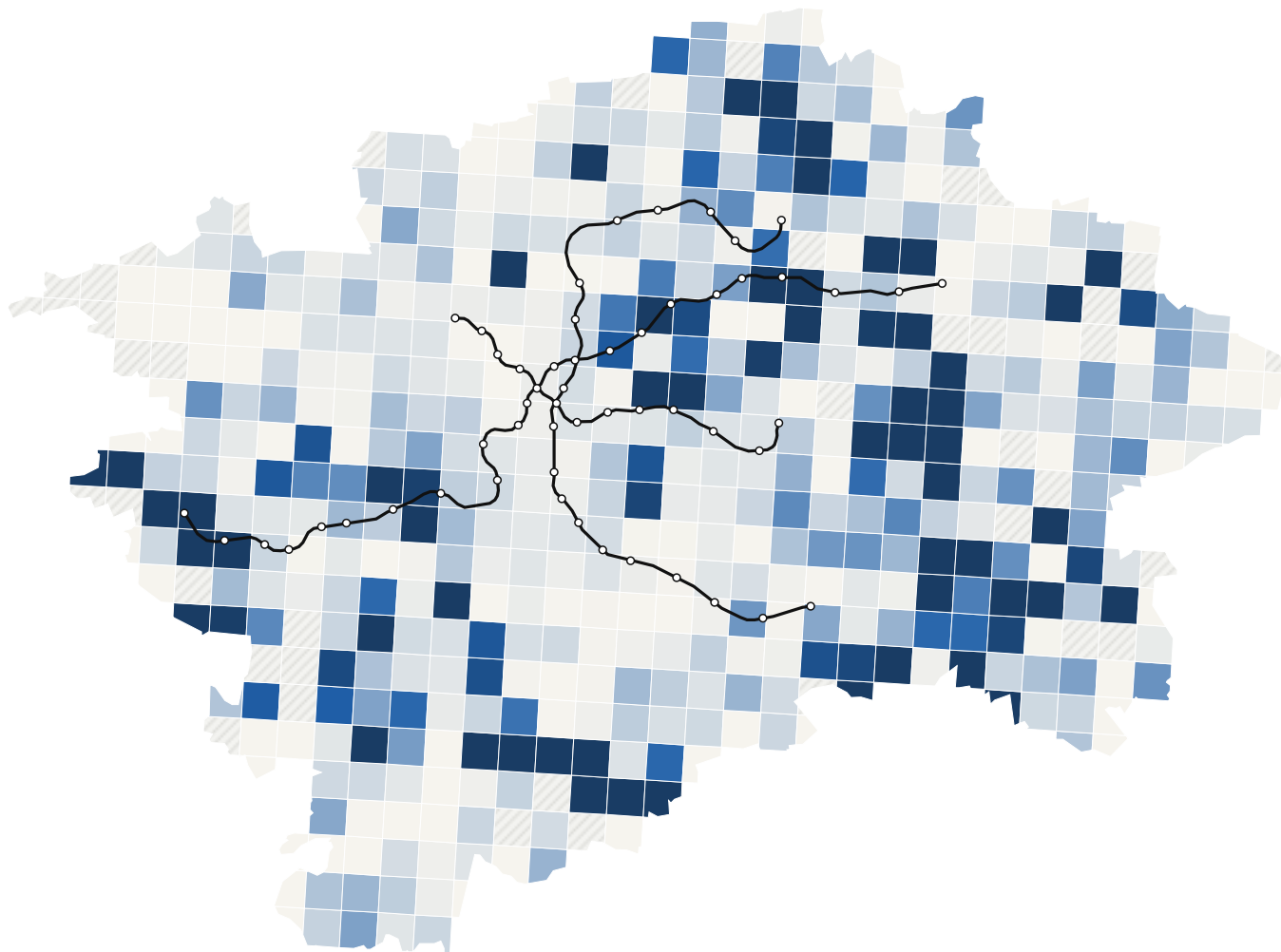
fig. 4 · Share of dwellings completed in 2024 by builder, in %; administrative counts, Vienna $n = 12\,478$, Warsaw $n = 14\,873$. Vienna: limited-profit housing associations (blue), public sector (black, 2.4 %), companies (light grey), private persons (grey, 6.8 %). Warsaw: municipal homes and public building societies (TBS, blue) against everyone else (light grey). Sources: Statistik Austria, Statistics Poland, accessed 28 September 2026.

In 2024 a quarter of Vienna's new homes, 3 107 of 12 478, were built by limited-profit housing associations, companies whose profits Austrian law restricts, and another 2.4 % by the public sector itself. In Warsaw the city built 96 homes that year and its public building societies none, 0.7 % of the year's 14 873. The Czech Statistical Office does not publish Prague's completions by builder, so for Prague the comparison stops at volume.

No yearly Austrian series by builder was found, but Poland records, for each of its 380 districts, the dwellings that municipalities, social building societies and developers begin each year, so the registered question concerns Poland only: whether non-market builders respond less to local price growth than developers. The point estimate goes the expected way, -0.56 (90 % interval -1.52 to $+0.39$, $p = 0.17$), with the 242 districts on common support acting like about 63 independent clusters, and the design had put power at the smallest difference of interest at 19 %. The relevance check fails, because developers' own starts do not visibly respond to local price growth either (0.09 , $p = 0.32$), which leaves no response for the non-market builders to be smaller than, and the result is labelled uninformative. Subsidised construction can partly replace what the market would have built^{34,35}, and the Polish test says nothing about what limited-profit associations would do in Prague.

New flats and distance to the metro in Prague

Land near a metro station is worth more, so an unconstrained market would build more densely there, although near-metro squares are also largely built-out estates with little land left to build on. The test counts flats completed between 2012 and census day in March 2021 in each of Prague's 524 one-kilometre squares, holding equal the distance from the centre and how many people already lived in the square in 2011.



0 200+ · flats per 1 000 residents · darkest: 200 or more

Flats completed 1 Jan 2012 to 25 Mar 2021 in Prague's 1 km squares; RÚIAN, GEOSTAT 2011.

no data

map · Flats completed in Prague's 1 km squares from 1 January 2012 to census day, 25 March 2021, per 1 000 residents of the square in 2011; hatched: nobody lived there in 2011. Above the map the number of new flats or the distance from the square's centre to the nearest station can be picked instead, and hovering over or tapping a square gives its numbers. Black: the metro and its stations as they ran in 2012, without the 2015 extension of line A. Source: ČÚZK RÚIAN building points (accessed 29 September 2026), GEOSTAT 2011 population grid, Census 2021 1 km grid.

● main estimate ● variant, $p < 0.05$ ● variant, $p \geq 0.05$ --- smallest difference of interest

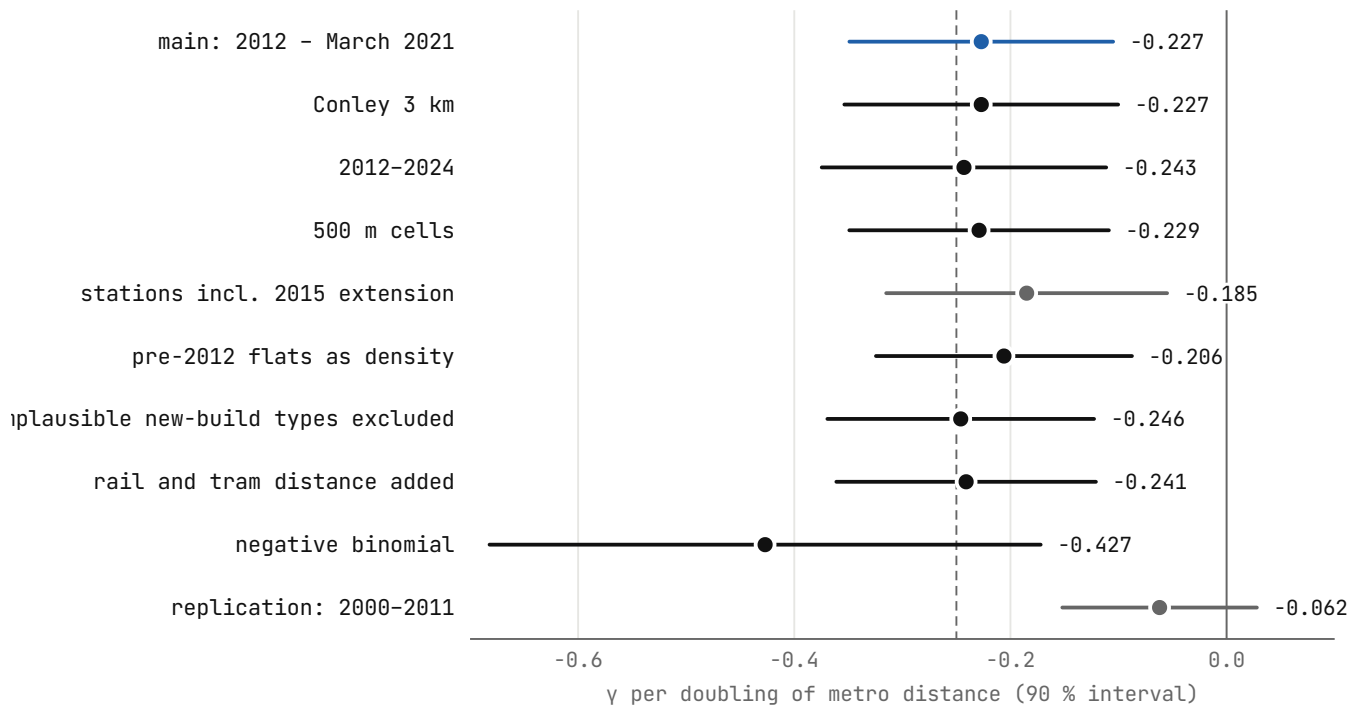


fig. 5 · Change in log new-flat density per doubling of the distance to the nearest metro station, 524 squares, with 90 % spatial (Conley) intervals; negative binomial with district-clustered intervals. Blue: the main estimate; black: variants whose decision p is below 0.05; grey: the rest. Grey dashed: the smallest effect the design cared about. Not drawn: clustering by the 57 city districts ($p = 0.015$). Source: ČÚZK RÚIAN, accessed 29 September 2026.

The estimate is $\gamma = -0.227$ (90 % Conley interval -0.349 to -0.105), which means that each doubling of the distance to a station goes with about 20 % fewer new flats per square kilometre. The registered decision takes the larger of two p -values, 0.0011 from the spatial (Conley) standard error and 0.0252 from the wild bootstrap over Prague's 22 administrative districts, which act like fewer than 14 independent clusters. Holm's second threshold is 0.025, so H_5 does not pass, and because the reverse test does not come close either, the registered label is indeterminate. The margin, 0.0002, is within the bootstrap's own simulation error of about 0.0016, and the test was not re-run. Even a passing result would have left the reason open, because squares near stations differ from others in the land left to build on and in what the plan allows there, and those layers could not be obtained in time.

Robustness

Result	Variant	Estimate
H2	Jihomoravský kraj, home of Brno, as the only comparison	+7.6 months
H2	Středočeský kraj counted as Prague	+10.3 months
H2	family houses added	+5.0 months, $p = 0.055$
H2	exponential lag; without the COVID months	a few bootstrap draws run off to unbounded lags; no information, counted as failures
H5	half-kilometre squares, 2012–2024, implausible types excluded, rail and tram added, 57 city districts	negative in every variant (fig. 5 and its caption)
H5	dropping any one district	–0.27 to –0.18
H5	2000–2011, before the window	–0.06, uncertain
H1	seven variants of the demand model (numerator, sample, covariates, weights)	45th of 116 to 138th of 208
H3	Warsaw, notified starts of works, not tested	–5.7, 4th of 376 districts

Specification curve	specifications	same sign	$p < 0.05$	robust by the registered rule
H2 · permit to house number	9	100 %	67 %	yes
H4 · builders, Poland	8	88 %	13 %	no
H5 · the metro	8	100 %	88 %	yes

Registered rule: robust if at least 75 % of specifications share the sign and at least 50 % reach $p < 0.05$.

Robustness bootstraps use 999 draws. Data: [housing_extended_robust.json](#).

The sign of the pipeline difference holds in every variant that was run, and the metro gradient stays negative in all of them, although in 2000–2011 it was about a quarter as steep. Against the demand model, which failed its relevance check, Prague stays out of both tails in every variant.

What changed after registration

- 29 September 2026, choices the design left open: census reference dates come from national documentation, and several are marked for verification; a metro region's core is its densest member; the Czech rebuilt numerator uses monthly regional completions, and the within-Czechia comparison is the raw rate, with no model; lags are the best of 10 starting values; twelve quarterly lags leave 27 quarters, not 35, and the fixed-b critical value is simulated; the Polish bootstrap is the unstudentised score version, its effective number of clusters uses an independence working covariance, and the Polish common support is 242 districts; rail and tram distances use today's network; half-kilometre squares take their parent square's density; robustness bootstraps use 999 draws.
- 29 September 2026, after estimation: H5 was decided at $p = 0.0252$ against 0.025 and not re-run. Prague's monthly noise was 3.3 times the signal, which triggers the registered rule that a failed pipeline test be written "underpowered"; the rule did not apply, because H2 passed. The rest of Czechia's 5-month lag and Prague's flat likelihood are flagged, and two pipeline variants had diverging bootstraps. Three relevance checks failed, and their results are labelled uninformative. Robustness variants for H4 drop district-builder cells with no dwellings at all, which the main frame did not have.
- 29 September 2026, after the fact-check: two registered checks omitted from the first runs, the metro test clustered by the 57 city districts and the Warsaw replication, were added; neither is in the family.
- Not run, because the inputs could not be obtained before the outcomes were joined: the land and planning layers for the metro question (Urban Atlas, heritage zones, flood plains, the 1999 plan and line D) and the analysis by basic settlement unit; Polish municipal construction times, because the statistics API refused connections; the stock of dwellings under construction, because the Prague series has not been published

since 2009; one price-index variant and the annual version of the price question; and an OECD source beside the Doing Business figure, which was not verified in time.

What this means

In official monthly counts of flats in new apartment buildings, completions follow permits with a best-fitting mean lag 11.1 months longer in Prague than in the rest of Czechia: supported ($p < 0.001$), the one passing registered test, with a one-sided lower bound of 6.9 months at Holm's threshold. The sign holds, but the profile likelihood allows a size from about 6 months (3 counting an isolated grid point) to more than 30 months. The gap may partly reflect reporting timing and project size; the design identifies no cause, so it does not show that procedure after the permit delays Prague's flats, or what shortening it would change. It excludes the time taken to obtain a permit, the step the 2021 building act addressed.

Prague completed 4.5 homes per 1 000 residents a year over 2015–2024; by the 2011–2021 census count, which in Czechia includes reconstructions, its metropolitan region built 5.28 against a European median of 3.55.

Whether that is low for its demand is uninformative: the demand model failed its relevance check. The metro gradient, -0.23 in log density per doubling of distance (90 % Conley interval -0.349 to -0.105), is indeterminate and describes an association only.

What this does not show

The European comparison is descriptive and establishes no cause, and its demand model failed its relevance check. The pipeline result starts at the permit, so it does not measure the time it takes to get one, and the difference it measures may partly be reporting timing and project size. Because prices respond to supply as well, the price questions measure responsiveness rather than an elasticity. The builders' test is Polish and says nothing about Prague. The metro gradient is an association, and near-station squares differ in land and planning in ways that were not measured.

The cadastre keeps only standing buildings and dates the building rather than the flats added to it; the census also dates buildings and, in Czechia, Spain and Croatia, counts their reconstruction; one-kilometre squares and fourteen regions are coarse units, and results can change with the choice of areal unit³⁶.

Reproduction

The scripts live in `tools/praha/`. Raw downloads are cached in `tools/data/praha4/` and `tools/data/praha4x/`, which are not committed, and are checked against `docs/research/prague-housing-files.sha256`. Each script is run with `uv` and lists its dependencies in its header; they are not yet pinned to versions. The registered estimates are rebuilt with `DATA=tools/data uv run --with pandas --with numpy --with scipy --with statsmodels --with pyarrow --with openpyxl --with shapely --with pyproj --with pyfixest --with arch python tools/praha/housing_extended.py`, and the scripts run in this order:

1. `cities.py`: the three-city series and the 2024 builder split, `cities.json`;

2. `housing_fetch.py`: raw files and their hash list;
3. `housing_data.py`: the design frames, with no outcome summarised;
4. `housing_power.py`: design-stage power, `prague-housing-power.json`;
5. `housing_extended.py`: the registered estimates, `housing_extended.json`;
6. `housing_robust.py`: robustness, `housing_extended_robust.json`;
7. `housing_addenda.py`: the checks added after the fact-check;
8. `figures_housing_extended.py`: the figures.

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Change log

- 28 September 2026: first published, with the three-city comparison and the 2024 builder split.
- 29 September 2026: extended with the registered study (version 1); after the fact-check, the metro test clustered by the 57 city districts and the Warsaw replication were added.
- 29 September 2026: version 2. Retitled from "Prague builds two homes for Vienna's three"; the kicker, deck and facts no longer present Prague as an under-builder against two high-building comparators, and the text states

that Prague's metropolitan region builds above the European median. "Pre-registered" replaced by the registration line. H2: the profile-likelihood region and the one-sided lower bound at Holm's threshold (6.9 months, from the bootstrap standard error in housing_extended.json) added, and the 11 months qualified as a difference in the recorded pipeline that may partly be reporting timing. H5 labelled indeterminate throughout. The price-response section folded into the note under the tests; the Polish builders' interval figure (fig. 5 of version 1) and the fact cards on Vienna, Warsaw and the non-profit share removed; the robustness ranks moved into a table. An unsupported sentence on Warsaw's price growth and a closing remark removed. References renumbered, one work per entry, with access dates and Eurostat table codes; the uncited reference to Parts 1-3 removed. The parking study is now Part 5. Question, data and reproduction sections added.

- 29 September 2026: added the section What this means.
 - Checked against editorial standard v1 on 29 September 2026.
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